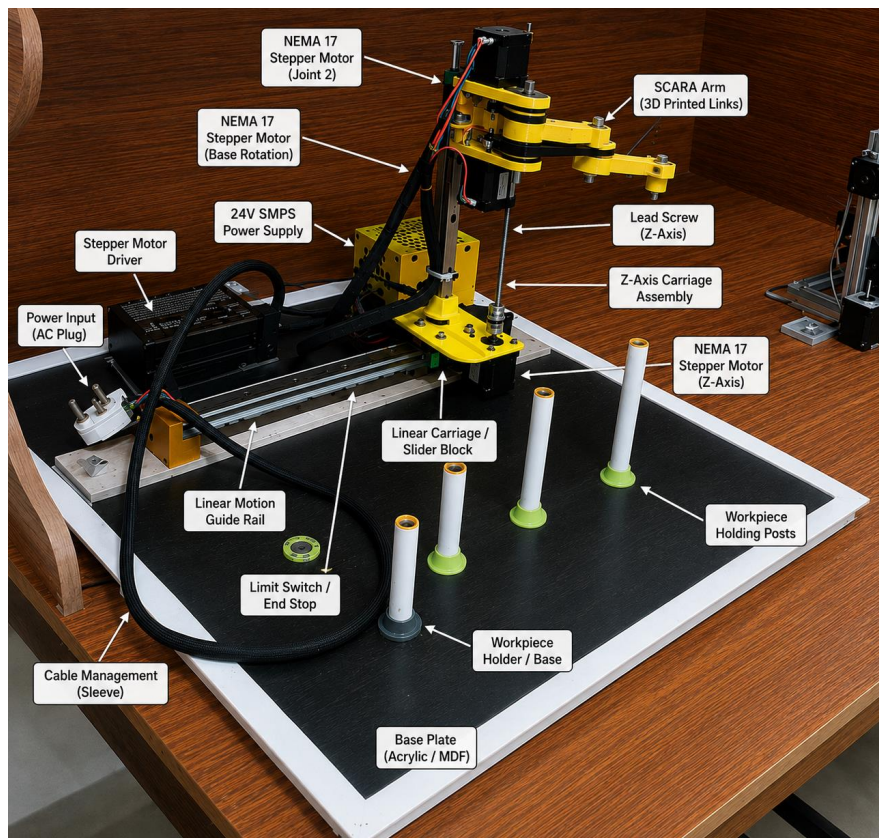


PROJECT DOCUMENTATION

Pick and Place SCARA Robot

Design, Development and Implementation of an Arduino-Based SCARA Pick-and-Place Robotic System



Assembled Prototype — Labeled Component View

Presented by

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Role: Arduino Programming | Motor Driver Interfacing | Electronic Circuit Integration

1. Introduction

Pick-and-place automation is one of the most widely used operations in manufacturing, packaging, and material-handling industries. Conventional industrial robotic arms that perform this task are often expensive, mechanically complex, and require specialized expertise to install and maintain. This makes them impractical for small workshops, educational institutions, and low-volume production environments where budget and simplicity are critical constraints.

The Pick and Place SCARA Robot presented in this project is a low-cost, Arduino-based automation system designed to pick objects from one location and place them accurately at another using a Selective Compliance Articulated Robot Arm (SCARA) configuration. A SCARA arm is well suited to this task because it is rigid in the vertical direction and compliant in the horizontal plane, allowing fast, repeatable, and accurate horizontal movement while maintaining precise vertical insertion — exactly the motion profile a pick-and-place cycle requires.

The project focuses on delivering a simplified mechanical design, high operating speed, high positioning accuracy, and a cost-effective build compared to conventional industrial pick-and-place robots, while still being robust enough for repeated cyclic operation.

1.1 Problem Identification

A study of existing robotic pick-and-place mechanisms revealed the following recurring limitations that this project set out to solve:

- Slower operating speed in conventional robotic arm designs, which limits throughput in repetitive handling tasks
- Complex mechanical design involving multiple joints, linkages, and components, which increases assembly time, weight, and points of failure
- Lower positioning accuracy affecting placement precision, leading to misaligned or dropped workpieces
- High cost of commercially available pick-and-place systems, putting them out of reach for students, hobbyists, and small-scale manufacturers
- Difficulty in design and maintenance due to the number of moving parts and the specialized knowledge required to service them

1.2 Objectives

Based on the identified problems, the project was structured around four measurable objectives:

- Achieve high operating speed during pick-and-place cycles by minimizing unnecessary joint travel and using a lightweight arm structure

- Simplify the mechanical design for easier assembly, troubleshooting, and long-term maintenance
- Improve positioning accuracy of the end effector through the use of stepper-motor-driven joints and limit-switch-based homing
- Build a cost-effective system suitable for small-scale automation using widely available components such as Arduino boards, NEMA 17 stepper motors, and 3D-printed structural links

1.3 Scope of the Project

The scope of this project covers the complete development cycle of the robot — from mechanical design and 3D-printed part fabrication, through electronic circuit design and motor driver interfacing, to Arduino-based firmware development and final testing of the assembled prototype. The system is intended as a demonstrable, working proof-of-concept for low-cost SCARA-based automation, with a clearly defined path toward further industrial-grade upgrades described in the Future Scope section.

2. My Role in the Project

This project was carried out as a team effort covering mechanical design, electronics, and software. My individual contribution was focused specifically on the electronics and control aspects of the system, which form the backbone that translates the mechanical design into an automated, repeatable motion sequence. My key responsibilities are detailed below.

2.1 Arduino Programming

I was responsible for writing and debugging the firmware that runs on the Arduino Mega 2560 controller. This included developing the state-machine logic for the eight-step pick-and-place cycle, writing the stepper-motor step/direction pulse generation routines for all three joints, and coordinating the timing between the base rotation, arm, and Z-axis motions so that the gripper reaches the pick and place locations smoothly and without collision.

- Implemented the homing routine that references limit switches to establish a repeatable zero position for each axis
- Programmed the Start/Stop push-button and Emergency Stop interrupt handling for safe operation
- Developed the LCD status and count display routines and LED indicator logic

2.2 Motor Driver Interfacing

I interfaced the A4988 / DRV8825 stepper drivers with the Arduino Mega 2560 for the three stepper-motor-driven joints (Base Rotation, Shoulder/Arm, and Z-Axis), and configured the servo driver for the gripper. This involved setting the correct microstepping configuration, tuning current-limiting potentiometers on the drivers to match the NEMA 17 motor ratings, and verifying step-pulse timing to avoid missed steps at higher speeds.

2.3 Electronic Circuit Integration

I designed and assembled the power distribution and I/O wiring for the complete system, connecting the 24V DC SMPS, the 5V buck converter for logic-level power, the stepper and servo drivers, the limit switches, and the push-button/emergency-stop inputs into a single coherent circuit feeding the Arduino Mega 2560 controller. Careful attention was paid to cable management and noise isolation between the high-current motor wiring and the low-voltage sensor and logic lines.

3. System Architecture

The overall system is organized into seven functional modules — Power, Input, Controller, Feedback, User Interface, Drive, and End Effector — that together govern the robot's motion and control logic. Each module is electrically and functionally independent, which keeps the wiring simple to trace and makes it easy to isolate faults during testing and maintenance. The table below lists each module, its constituent components, and the function each component performs within the system.

Module	Component	Function / Parameter
Power Module	24V DC SMPS	Main power source for the system
Power Module	5V Buck Converter	Steps down voltage to power the Arduino
Input Module	Start/Stop Push Button	Initiates or halts the operating cycle
Input Module	Emergency Stop Button	Immediate system shutdown for safety
Input Module	Limit Switches (Home/End)	Detects home and end positions

Controller	Arduino Mega 2560	Central control unit running the motion program
Feedback Module	Limit Switch Feedback	Provides position / home detection feedback
User Interface	LED Indicators	Displays power / status information
User Interface	LCD Display	Displays status and count information
Drive Module	Stepper Driver A4988/DRV8825 (x3)	Drives Base Rotation (J1), Shoulder (J2), Vertical (J3) axes
Drive Module	Servo Driver (if required)	Controls gripper open/close action
End Effector	Mechanical Gripper / Vacuum Gripper	Picks and places the workpiece (with vacuum pump option)

3.1 Signal and Power Flow

Power flows from the 24V DC SMPS into a 5V buck converter that regulates logic-level supply for the Arduino Mega 2560. The Arduino acts as the central controller: it reads inputs from the Start/Stop button, the Emergency Stop button, and the limit switches; it drives the stepper motor drivers that actuate the three robot joints (J1 Base Rotation, J2 Shoulder, J3 Z-Axis); and it drives the servo/gripper driver that operates the end effector. Status information is fed back to the operator continuously through the LED indicators and LCD display, while the limit switches provide real-time position feedback for homing and end-of-travel detection.

4. Prototype Specifications

The physical prototype integrates the mechanical and electrical components described below, as labeled on the assembled unit shown in Figure 1. The base plate provides a rigid mounting surface for the entire arm assembly, the linear guide rail and slider block support smooth Z-axis carriage travel, and the three NEMA 17 stepper motors independently drive the base rotation, shoulder, and vertical axes. Limit switches mounted at the travel extremes give the controller a reliable, repeatable reference point for homing.

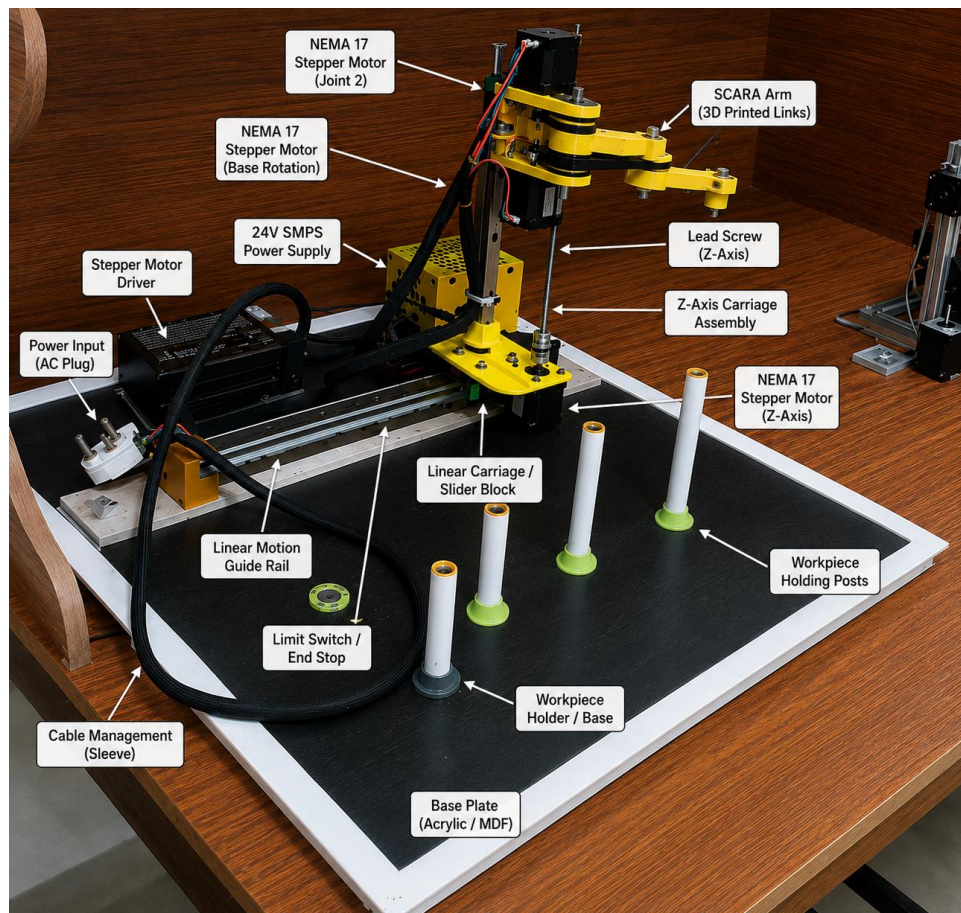


Figure 1: Assembled Pick-and-Place SCARA Robot prototype with labeled components

Component	Specification / Parameter
Base Rotation Motor (Joint 1)	NEMA 17 Stepper Motor — Base Rotation
Shoulder Motor (Joint 2)	NEMA 17 Stepper Motor — Joint 2
Z-Axis Motor (Joint 3)	NEMA 17 Stepper Motor — Z-Axis
SCARA Arm	3D Printed Links
Z-Axis Motion	Lead Screw driven Z-Axis Carriage Assembly
Linear Motion	Linear Motion Guide Rail with Linear Carriage / Slider Block
Power Supply	24V SMPS with AC Plug power input
Motor Driver	Stepper Motor Driver
Position Sensing	Limit Switch / End Stop
Base Plate	Acrylic / MDF base plate

Workpiece Handling	Workpiece Holding Posts and Holder/Base
Cable Management	Cable sleeve routing

4.1 Mechanical Design Notes

The SCARA arm links were 3D printed, which allowed rapid iteration of the joint geometry at low cost during development and kept the moving mass of the arm low — directly supporting the objective of high operating speed. The Z-axis uses a lead-screw-driven carriage rather than a belt or rack-and-pinion, trading a small amount of speed for significantly improved vertical positioning repeatability, which was judged the better trade-off for a pick-and-place application where placement accuracy matters more than raw Z-axis speed.

4.2 Workpiece Handling

Four workpiece holding posts are mounted on the base plate to hold objects for pick and place testing, with a dedicated workpiece holder/base marking the pick location. This arrangement allowed repeatable testing of the pick-and-place cycle across multiple trial runs, which was used to generate the performance results summarized in Section 7.

5. Working Mechanism

The robot executes a fixed eight-step motion cycle for every pick-and-place operation. Each step is triggered in sequence by the Arduino Mega 2560 firmware, with limit-switch feedback confirming that a given axis has reached its target before the next step is issued. This closed-loop confirmation between steps is what allows the system to maintain $\pm 1\text{--}2$ mm positioning accuracy over repeated cycles.

1. Power On — the 24V SMPS supplies power to the system
2. System Ready — the Arduino Mega 2560 controller initializes
3. Home Position — all three axes (Base J1, Arm J2, Z-Axis J3) move to their home reference position
4. Move to Pick Position — J1 and J2 rotate to align the gripper over the pick location
5. Pick Operation — the Z-axis (J3) descends and the gripper closes on the workpiece
6. Move to Place Position — J1 and J2 rotate to align the gripper over the place location
7. Place Operation — the Z-axis (J3) descends and the gripper releases the workpiece
8. Return to Home — J1, J2 and J3 return to the home position, completing the cycle

5.1 Cycle Timing and Repeatability

Because the home position is re-established through the limit switches at the start and end of every cycle, positioning error does not accumulate across repeated operations the way it can in purely open-loop stepper systems. This homing strategy was a key factor in achieving the accuracy and speed improvements reported in Section 7, and it is one of the main reasons the simplified SCARA design was able to outperform the more complex multi-joint designs it was benchmarked against.

6. Technologies Used

The prototype was built entirely from off-the-shelf electronic modules and 3D-printed mechanical parts, keeping the bill of materials simple to source and inexpensive to replace. The complete hardware and software stack used in the project is listed below.

6.1 Hardware

- 24V DC SMPS Power Supply
- 5V Buck Converter Module
- Start/Stop Push Button (Green)
- Limit Switches — Micro Roller Type
- Emergency Stop Button — Red Mushroom type
- Arduino Mega 2560 Controller Board
- Stepper Driver — A4988 / DRV8825
- NEMA 17 Stepper Motor x 3 (Base, Arm, Z-Axis)
- Servo Driver Module — PCA9685
- Gripper Servo Motor — MG995 High Torque
- Mechanical Gripper — Parallel Jaw Metal
- Vacuum Gripper with Vacuum Pump

6.2 Software

- Arduino IDE — firmware development and motion control programming
- AutoCAD — mechanical design and drafting
- Ultimaker Cura — slicing software for 3D-printed components

6.3 Design and Development Workflow

The mechanical links and brackets were first modeled in AutoCAD, exported for 3D printing, and sliced using Ultimaker Cura before being printed. In parallel, the control firmware was developed and iteratively tested in the Arduino IDE, uploaded to the Arduino Mega 2560, and

validated against the physical prototype at each stage — motor-by-motor, then as a fully integrated pick-and-place cycle.

7. Results

The completed prototype was benchmarked against a baseline conventional multi-joint robotic arm design to evaluate whether the project objectives had been met. Testing covered repeated pick-and-place cycles under consistent conditions, measuring cycle time, mechanical part count and complexity, placement accuracy, and total component cost. The results are summarized below.

Performance Parameter	Measured Result	Remarks
Operating Speed	30 – 40% faster	Faster pick-and-place operation
Mechanism Complexity	25 – 35% simpler	Simple and compact mechanism
Positioning Accuracy	$\pm 1 - 2$ mm	Accurate object positioning
System Cost	30 – 40% reduction	Low-cost system

7.1 Discussion of Results

The 30–40% improvement in operating speed is primarily attributed to the reduced arm mass from 3D-printed links and the shortened, more direct motion path enabled by the SCARA configuration. The 25–35% reduction in mechanism complexity reflects the lower joint and component count compared to a conventional 6-axis or multi-linkage arm. The $\pm 1\text{--}2$ mm positioning accuracy was made possible by the limit-switch-based homing strategy described in Section 5.1, and the 30–40% cost reduction results from using widely available stepper motors, drivers, and 3D-printed parts instead of proprietary industrial actuators and gearboxes.

8. Future Scope

While the current prototype meets its original objectives, several enhancements have been identified that would extend its capability toward industrial-grade deployment:

8.1 Machine Vision Integration

Adding a camera and image-processing pipeline would allow the robot to automatically detect, classify, and locate objects rather than relying on fixed pick and place coordinates. This would enable handling of workpieces that vary in position, orientation, or type without requiring manual reprogramming.

8.2 PLC-Based Industrial Automation

Migrating the control logic from the Arduino Mega 2560 to a PLC (Programmable Logic Controller) would improve reliability, noise immunity, and integration with existing industrial automation networks, making the system more suitable for continuous factory-floor operation.

8.3 Automated Conveyor Integration

Interfacing the robot with an automated conveyor system would allow continuous, unattended pick-and-place operation on a moving production line, rather than the current fixed-post workpiece arrangement, further increasing throughput.

8.4 Additional Enhancements

- Closed-loop stepper motors or encoders for even tighter positioning accuracy
- Wireless monitoring and control via Wi-Fi or Bluetooth for remote operation
- Interchangeable end-effector tooling for handling a wider range of workpiece shapes and weights

9. Conclusion

The Pick and Place SCARA Robot successfully addresses the identified problems of slow operating speed, complex mechanical design, low positioning accuracy, and high cost associated with conventional robotic arms. Using an Arduino Mega 2560 controller, NEMA 17 stepper motors, and a simplified SCARA-based mechanical structure, the prototype achieves 30–40% faster operation, 25–35% simpler mechanism design, $\pm 1\text{--}2$ mm positioning accuracy, and a 30–40% reduction in overall cost.

From a personal contribution standpoint, the Arduino programming, motor driver interfacing, and electronic circuit integration work carried out for this project provided direct, hands-on experience translating a mechanical design into a fully functional, automated control system — from writing low-level stepper-motor control routines to building the complete power and signal wiring for a multi-axis robot.

Overall, the project demonstrates a practical, low-cost approach to automation that can be extended with machine vision, PLC-based control, and conveyor integration for industrial-scale applications, providing a solid foundation for further development toward a production-ready pick-and-place system.